Earth–Sun Force Balance with $r_1 \cdot r_2$

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1. The balance condition

The modified law gives the real Newtonian force only when $r_1 \cdot r_2 = r^2$, so r is the geometric mean of r_1 and r_2.

Newton's law needs the gravitational constant $G = 6.674 \times 10^{-11}$ N·m²/kg². Labels follow the sketch: M_1 = Earth (small body), M_2 = Sun (large body).

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F = \frac{G\,m_1 m_2}{r^2} \qquad F' = \frac{G\,m_1 m_2}{r_1 r_2}
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Setting $F' = F$ gives the balance condition:

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r_1 \, r_2 = r^2 \quad\Longleftrightarrow\quad r = \sqrt{r_1 r_2}
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Every pair on the hyperbola $r_2 = r^2/r_1$ balances. If $r_1 = k \cdot r$, then $r_2 = r/k$: stretching one distance shrinks the other by the same factor.

Input	Value
m_1 (Earth)	5.972×10^{24} kg
m_2 (Sun)	1.989×10^{30} kg
r (Earth–Sun distance)	1.496×10^{11} m (1 AU)

2. Linking r_1 and r_2 to the masses

A symmetric split, where each body contributes an equal share, fixes the pair: $r_1/r_2 = m_1/m_2$.

The balance condition alone allows infinitely many pairs. Writing the force as a product of two "mass per distance" factors and making them equal adds the second rule:

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F = G\left(\frac{m_1}{r_1}\right)\left(\frac{m_2}{r_2}\right), \qquad
\frac{m_1}{r_1} = \frac{m_2}{r_2} \;\Rightarrow\; \frac{r_1}{r_2} =
\frac{m_1}{m_2}
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Solving this with $r_1 \cdot r_2 = r^2$:

$$r_1 = r\sqrt{\frac{m_1}{m_2}} \qquad r_2 = r\sqrt{\frac{m_2}{m_1}}$$

Each factor then equals $\sqrt{(m_1 m_2)}/r$, so the two bodies contribute equally to the force.

3. Earth–Sun numbers

For Earth and the Sun the split gives $r_1 \approx 259{,}000$ km and $r_2 \approx 577$ AU, and the force is 3.54×10^{22} N.

Quantity	Value	Note
m_1/m_2	3.003×10^{-6}	Earth is ~1/333,000 of the Sun
$\sqrt{(m_1/m_2)}$	1.733×10^{-3}	scale factor k
r_1	2.59×10^{8} m	~259,000 km, about 2/3 of the Earth–Moon distance
r_2	8.63×10^{13} m	~577 AU
$r_1 \cdot r_2$	2.238×10^{22} m²	equals r^2 ✓
F	3.54×10^{22} N	same as standard Newton

$$F = \frac{6.674\times10^{-11} \times 5.972\times10^{24} \times 1.989\times10^{30}}{2.238\times10^{22}} \approx 3.54\times10^{22}\ \text{N}$$

4. When Earth stays in its orbit

Earth keeps its circular orbit at 29.8 km/s only when $r_1 \cdot r_2 = r^2$; any other product changes the orbit.

Gravity must supply exactly the centripetal force, and the centripetal side uses the real distance r:

$$\frac{G\,m_1\, m_2}{r_1\, r_2} = \frac{m_1\, v^2}{r} \quad\Rightarrow\quad v = \sqrt{\frac{G\,m_2\,r}{r_1\, r_2}}$$

Case	Force	What Earth does
$r_1 \cdot r_2 = r^2$	Exactly Newton's	Circular orbit, $v = \sqrt{(G \cdot m_2/r)} = 29.8$ km/s
$r_1 \cdot r_2 < r^2$	Too strong	Falls inward onto a tighter ellipse
$r_1 \cdot r_2 > r^2$	Too weak	Drifts outward; far enough off, it escapes

The orbit fixes the product $r_1 \cdot r_2$; the masses only decide how that product is split between r_1 and r_2.

5. A split that fails: distances to the barycenter

Using each body's distance to the shared centre of mass makes the force about 333,000 times too strong.

- $r_1 = r \cdot m_2/(m_1+m_2) \approx 1.496 \times 10^{11}$ m (Earth to barycenter)
- $r_2 = r \cdot m_1/(m_1+m_2) \approx 449$ km (Sun to barycenter)

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r_1 r_2 = r^2\,\frac{m_1 m_2}{(m_1+m_2)^2} \approx 3\times10^{-6}\,r^2
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Summary

Condition	Meaning
$r_1 \cdot r_2 = r^2$	Required for the real force and a stable orbit
$r_1/r_2 = m_1/m_2$	Mass-weighted split with equal shares
$r_1 \approx 2.59 \times 10^8$ m, $r_2 \approx 8.63 \times 10^{13}$ m	Earth–Sun balance point
$F \approx 3.54 \times 10^{22}$ N, $v \approx 29.8$ km/s	Same as standard Newton